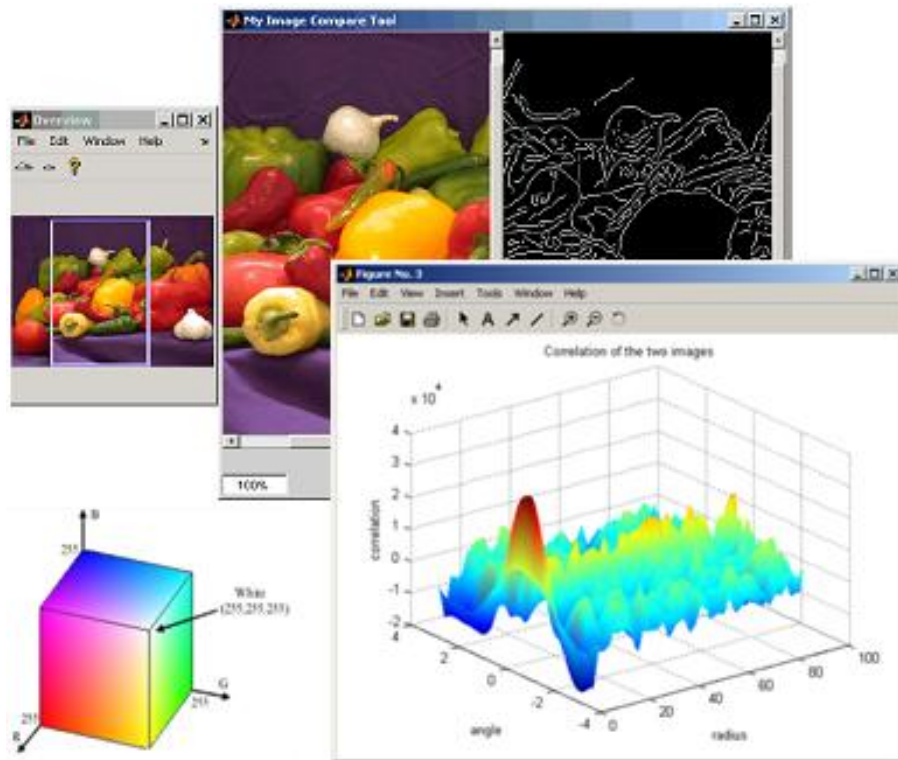


Digital Image Processing

LECTURE 8: IMAGE ENHANCEMENT



Dr. Ajay Kumar Mahato
Assistant Professor
ECE Department
GLA University Mathura



Image Enhancement

Image Enhancement

Image Enhancement is the process of **enhancing the quality** of a given image for analysis.

Image Enhancement

❑ Reasons for Image Enhancement:

- The aim of this process is to improve the quality of the image So that the image analysis is accurate, leading to an improvement in the reliability of the image processing applications.
- Making images more visually appealing
- Highlighting interesting details in images
- Removing noise from images

Image Enhancement

☐ **Images may suffer from the following degradations:**

- Poor contrast due to poor illumination
- Electronic sensor noise or atmospheric disturbances
- Aliasing effects due to inadequate sampling
- Finite aperture effects or motion

Image Enhancement

□ Applications of Image Enhancement:

- Contrast enhancement
- Gray scale transformation
- Photometric calibration
- Display calibration
- Contour lines

Image Enhancement

❑ Classification of Image Enhancement Techniques:

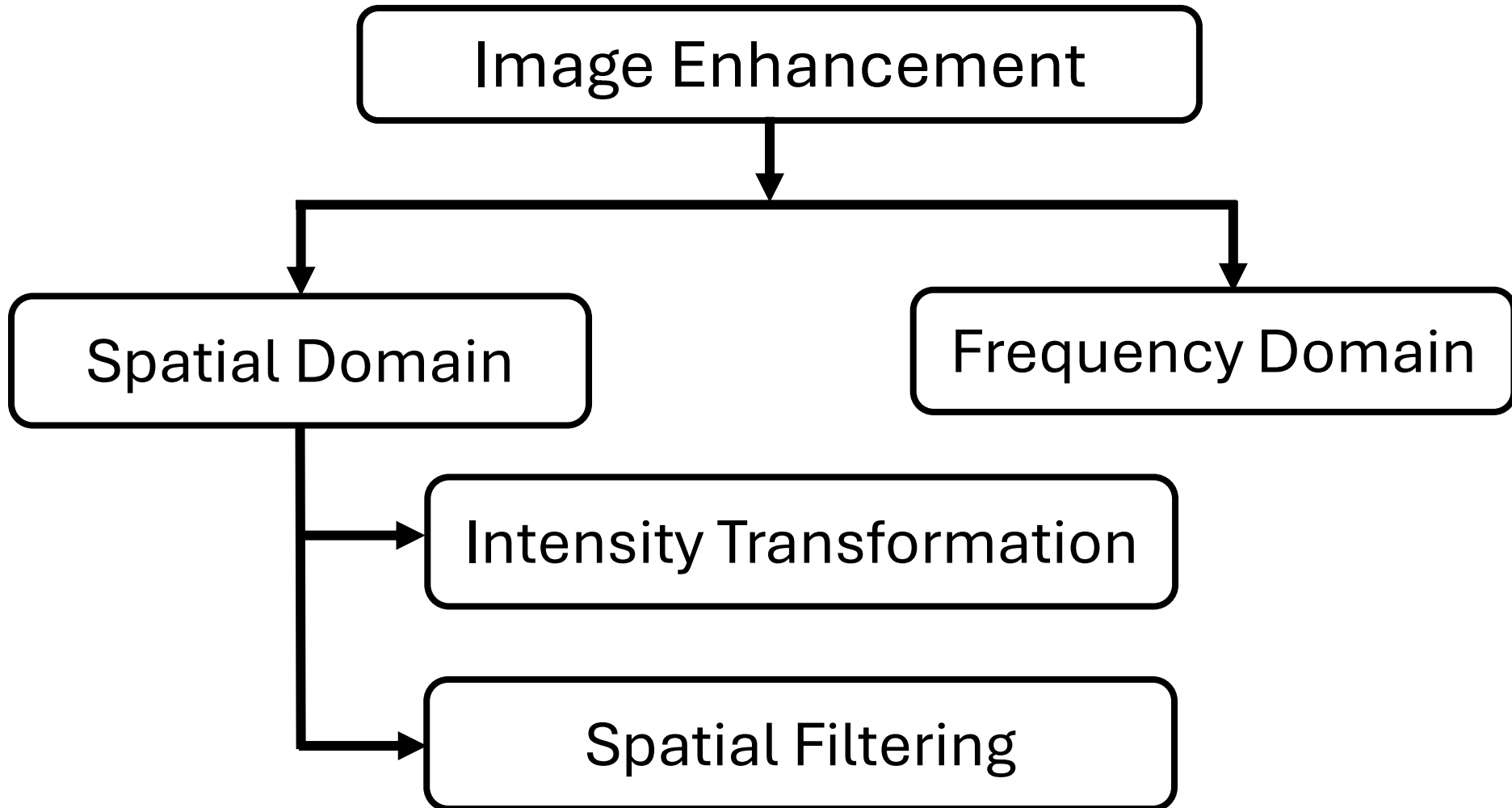


Image Enhancement

□ Spatial Domain:

- The term spatial domain refers to the image plane itself, and image processing methods in this category are based on direct manipulation of pixels in an image.

- Two principal categories:

- **Intensity Transformation:** Intensity Transformation operates on single pixels of an image for tasks.

Examples: Contrast Stretching, and Image thresholding

- **Spatial Filtering:** Spatial filtering performs operations on the neighbourhood of every pixel in an image.

Examples: Image smoothing, image sharpening, Image averaging

Intensity Transformation

Intensity Transformation: Intensity Transformation operates on single pixels of an image for tasks.

$$S = T(r)$$

where,

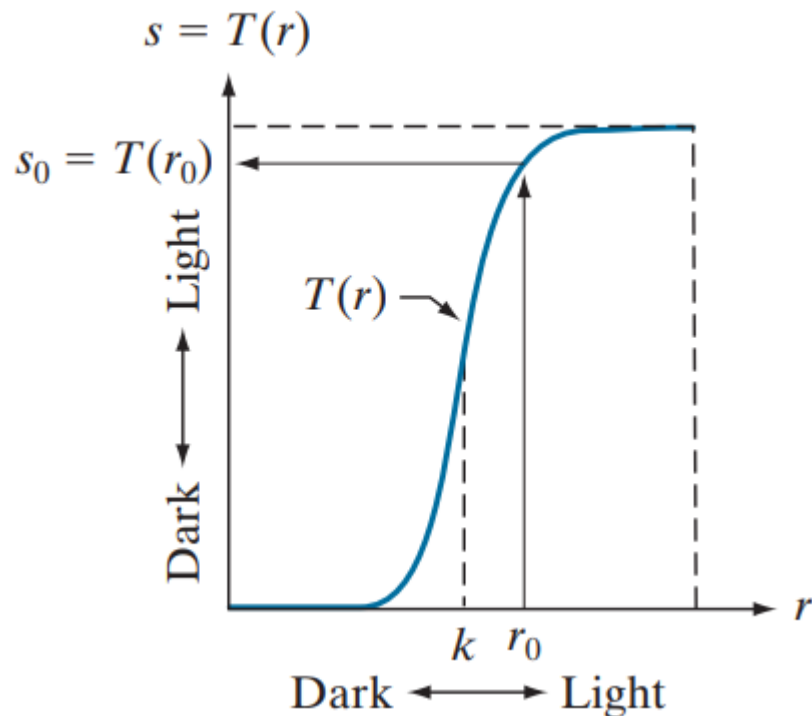
r : be the pixel input intensity before transformation

S : be the pixel output intensity after transformation

T : be the transform operator

Intensity Transformation

Intensity Transformation: Intensity Transformation operates on single pixels of an image for tasks.



Contrast Stretching

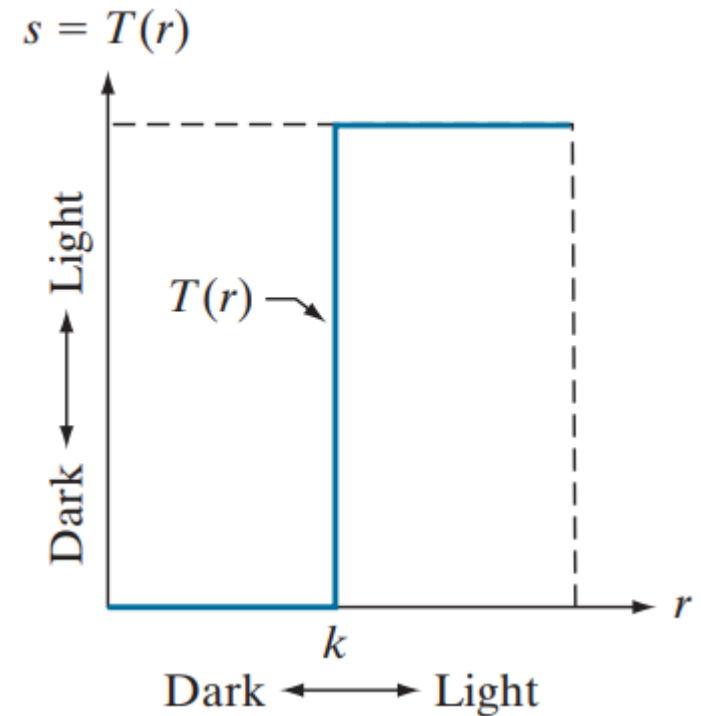


Image Thresholding

Intensity Transformation

❖ Different Methods of Intensity Transformation:

- ☐ Point Transformation

- ☐ Linear transformations

- Image negative
- Image Identity

- ☐ Non-linear transformations

- Logarithmic transformation
- Inverse Log transformation
- Power Law (Exponential) transformation

Intensity Transformation

☐ Different Methods of Intensity Transformation:

☐ Piecewise-linear transformations

- Contrast stretching
- Gray-level slicing
- Bit plane slicing

☐ Histogram

Point Transformation

Intensity Transformation

□ Point Transformation:

$$S = T(r)$$

where,

r : be the pixel input intensity before transformation

S : be the pixel output intensity after transformation

T : is a look up table.

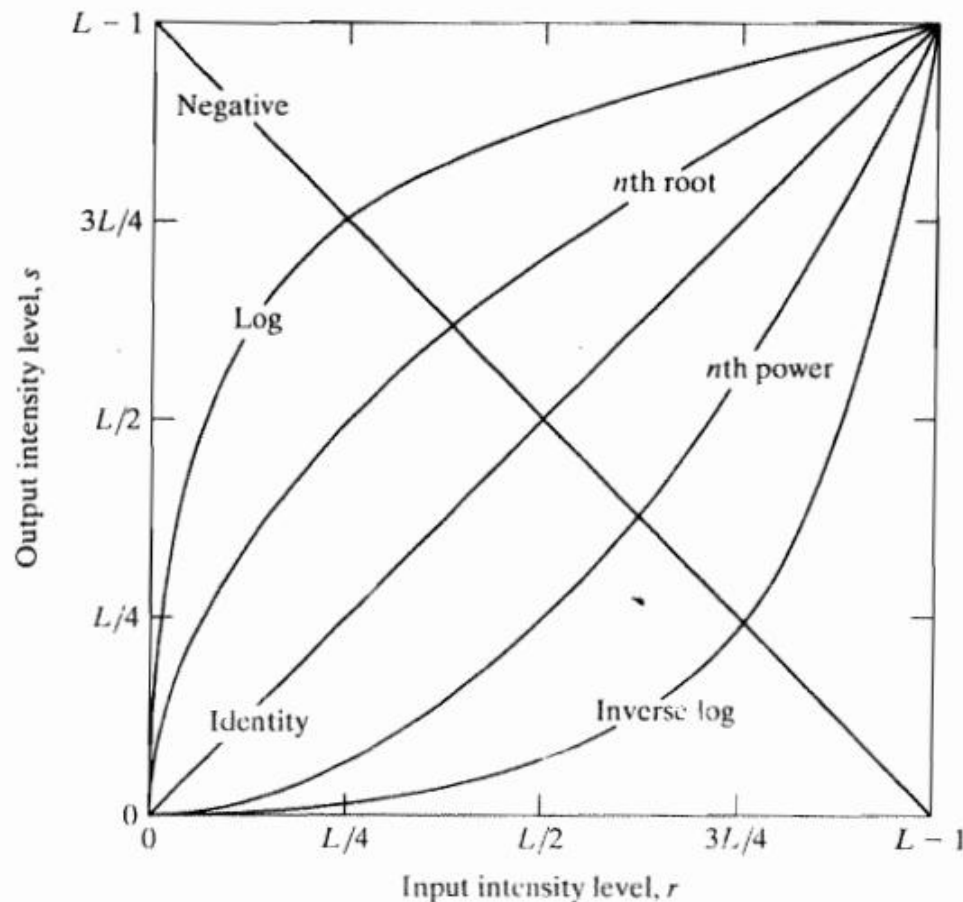
- Because we deal with digital quantities, values of an intensity transformation function are typically stored in a table, and the mapping from r to s are implemented via **table lookup**.
- For an 8-bit image, a lookup table containing the values of T will have 256 entries

Linear Transformation

Intensity Transformation

Linear Transformation:

- Image Negation: $S = L-1-r$
- Image Identity: $S = r$



Intensity Transformation

☐ Image Negation (on Gray Image):

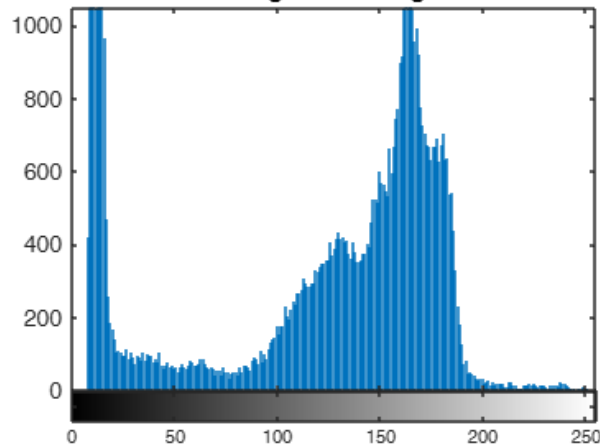
Original Image A



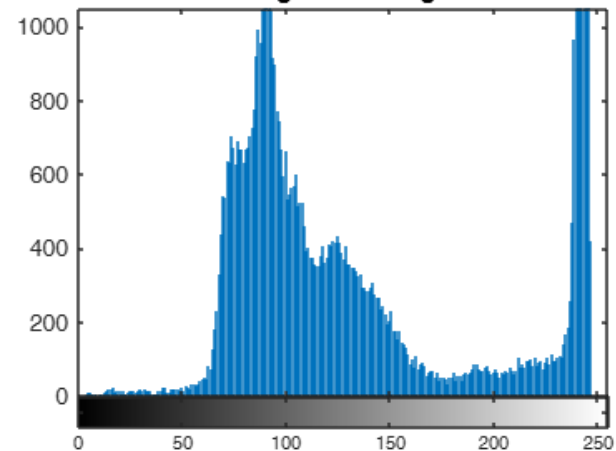
Complement of Image A



Histogram of Image A



Histogram of Image B



Intensity Transformation

❑ Image Negation (on RGB Image):

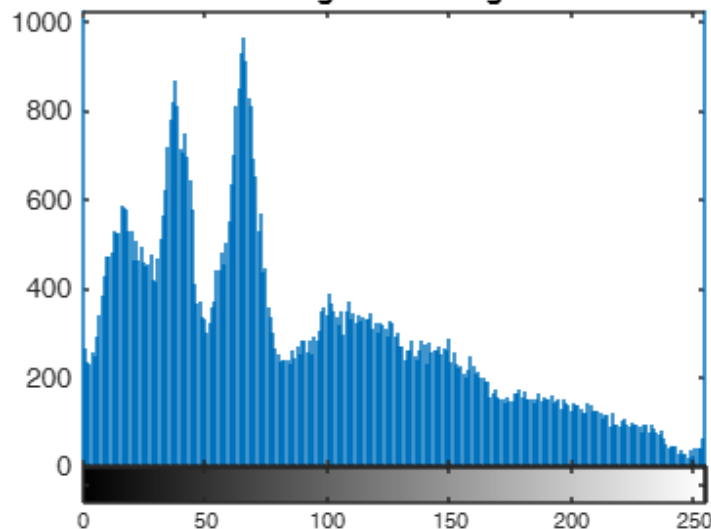
Original Image A



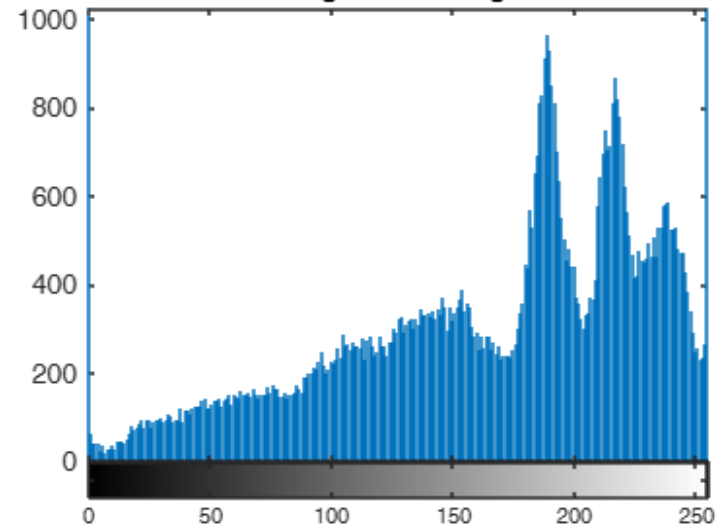
Complement of Image A



Histogram of Image A



Histogram of Image B



Numerical Examples

Ques 1: Consider the following 3×3 gray scale image

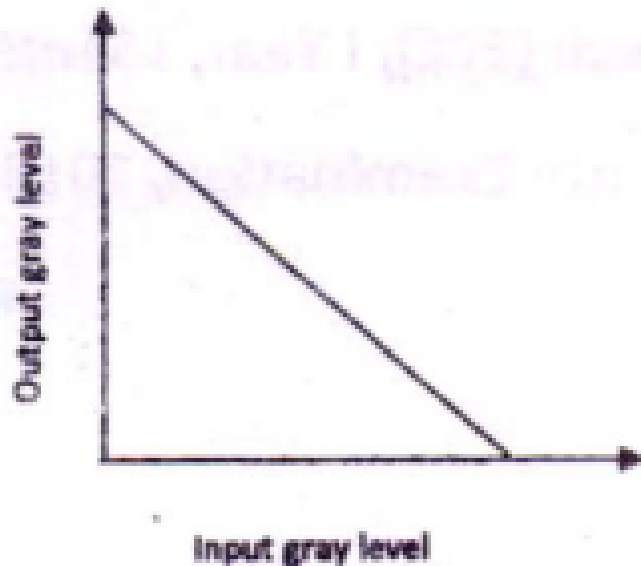
1	2	3
0	1	1
2	2	3

Obtain the image negative.

Numerical Examples

Ques 2: Name the transformation shown in figure and obtain that transform of the following image.

GLA University, Mid Sem Examination, Even Semester 2023-24



900	205	105
141	252	99
201	650	76

Non-Linear Transformation

Intensity Transformation

❑ **Non-linear (log) Transformation:**

$$S = c \log(1+r)$$

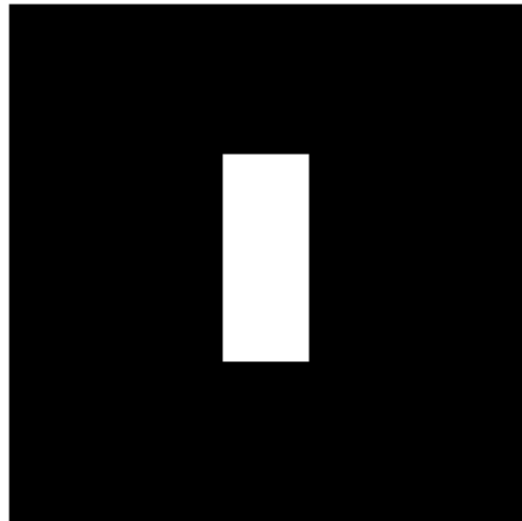
c is a constant

Applications: Used to expand the values of dark pixels in an image while compressing the higher-level values

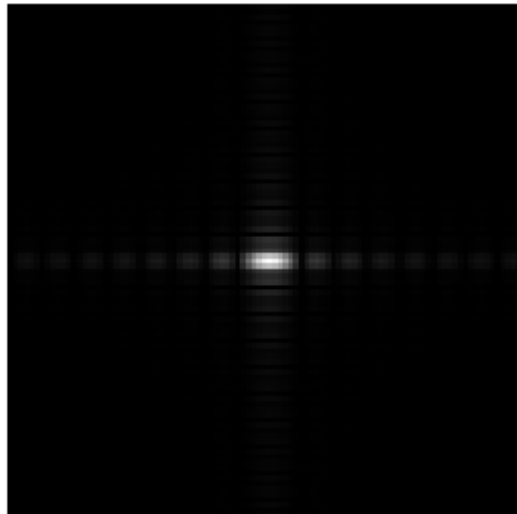
Intensity Transformation

☐ Non-linear (log) Transformation:

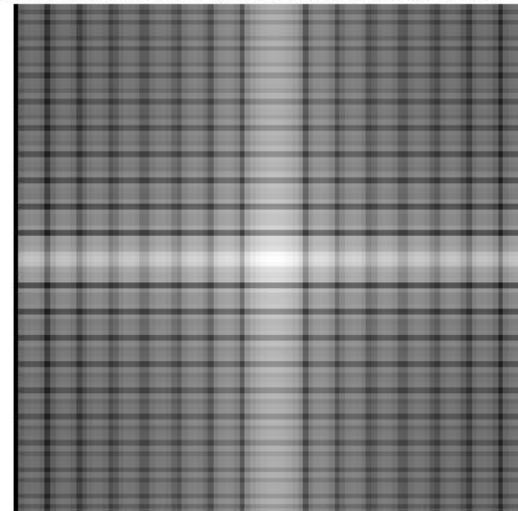
Gray Image B



Centered Spectrum F_c



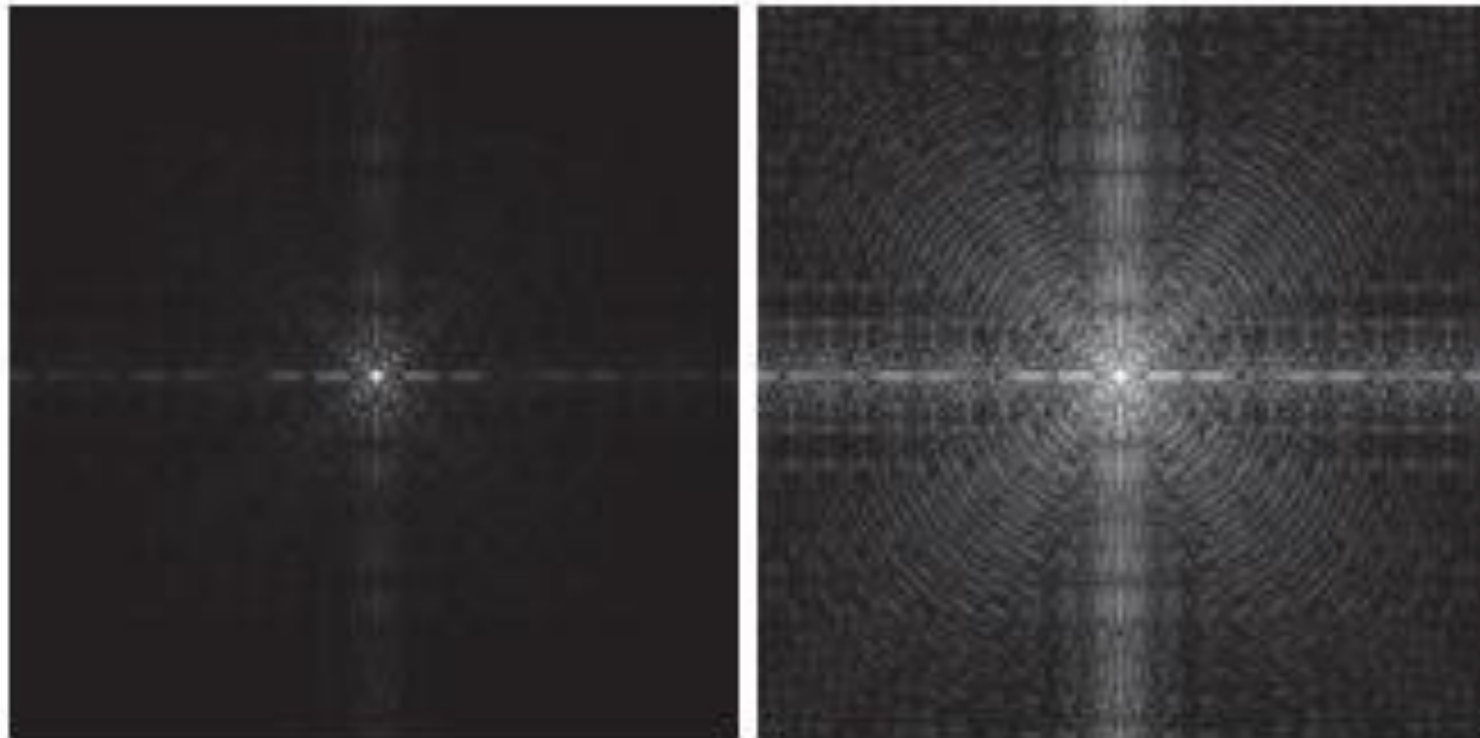
Spectrum Enhancement by Log Transformation



Intensity Transformation

❑ Non-linear (log) Transformation:

a b
FIGURE 3.5
(a) Fourier spectrum displayed as a grayscale image.
(b) Result of applying the log transformation in Eq. (3-4) with $c = 1$. Both images are scaled to the range $[0, 255]$.



Intensity Transformation

❑ Non-Linear (Inverse Log or Exponential) Transformation:

- This is the reverse of the logarithmic function and is given as follows:

$$g(x, y) = e^{f(x, y)}$$

- The modified exponential function is given as follows:

$$g(x, y) = c \times \left[(1 + \alpha)^{f(x, y)} - 1 \right]$$

- Here c , is the scaling factor. It can be observed that, if $f(x, y) = 0$ then $g(x, y) = 0$, thus, showing that one is subtracted as an offset.
- This function is used to enhance the details in high value regions of the image and decreases the dynamic range in low-value regions.

Intensity Transformation

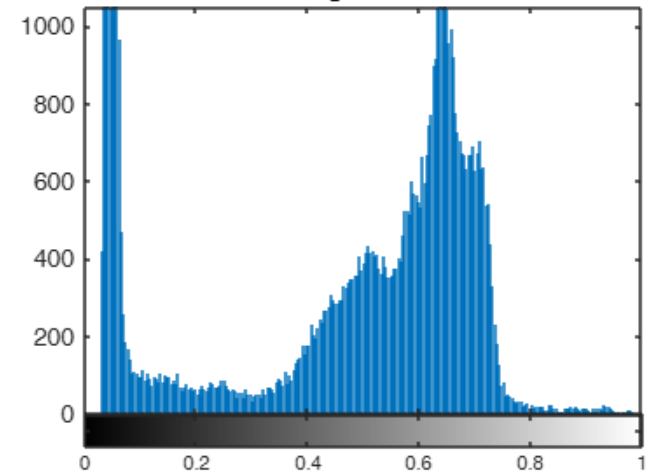
❑ Non-Linear (Inverse Log or Exponential) Transformation:

$$g(x, y) = C * e^{f(x, y)}$$

Double Image Ad



Histogram of Ad



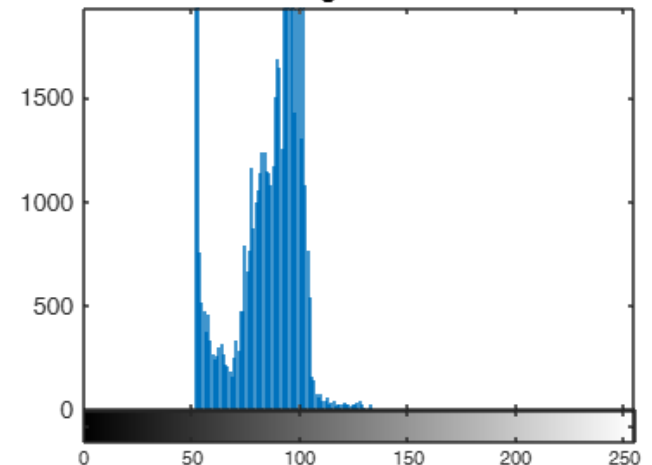
Original Image A



out=50.*exp(Ad)



Histogram of out



Intensity Transformation

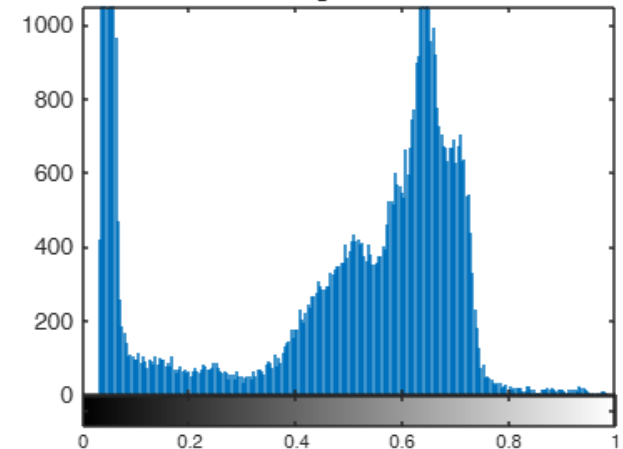
□ Non-Linear (Inverse Log or Exponential) Transformation:

$$g(x, y) = c \times [(1 + \alpha)^{f(x, y)} - 1]$$

Double Image Ad



Histogram of Ad



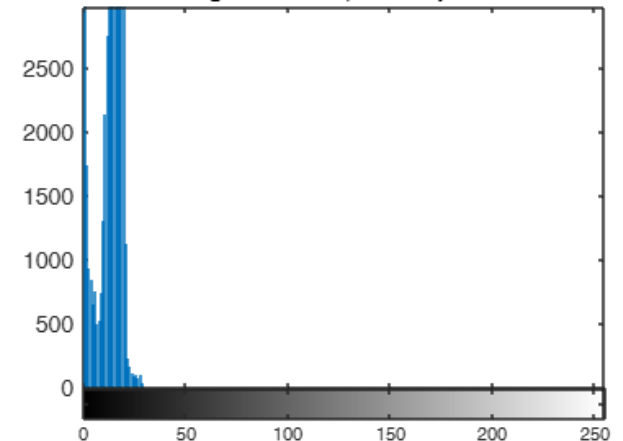
Original Image A



`out=50.*(((1+alpha).^(Ad)) -1);`



Histogram of out, with alpha = 0.0



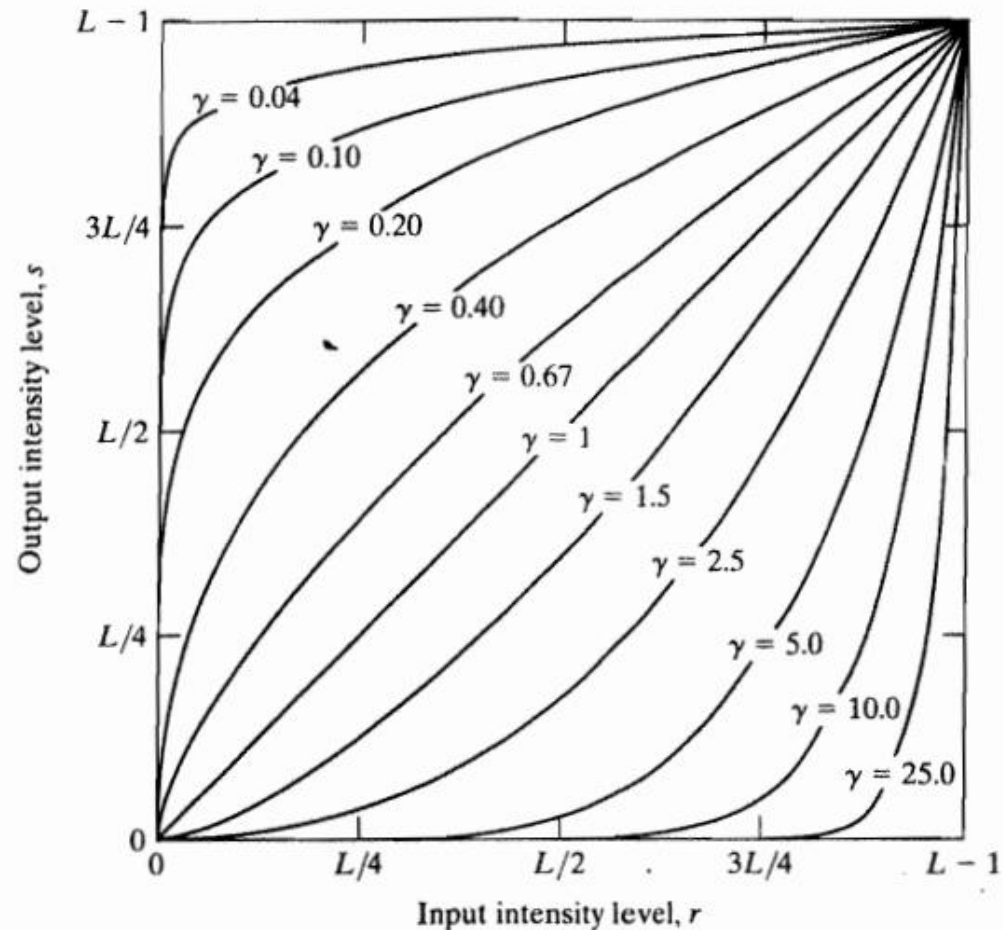
Power-Law Transformation

Intensity Transformation

❑ Power-Law Transformation:

$$S = c.r^\gamma$$

c and γ are positive constants



Intensity Transformation

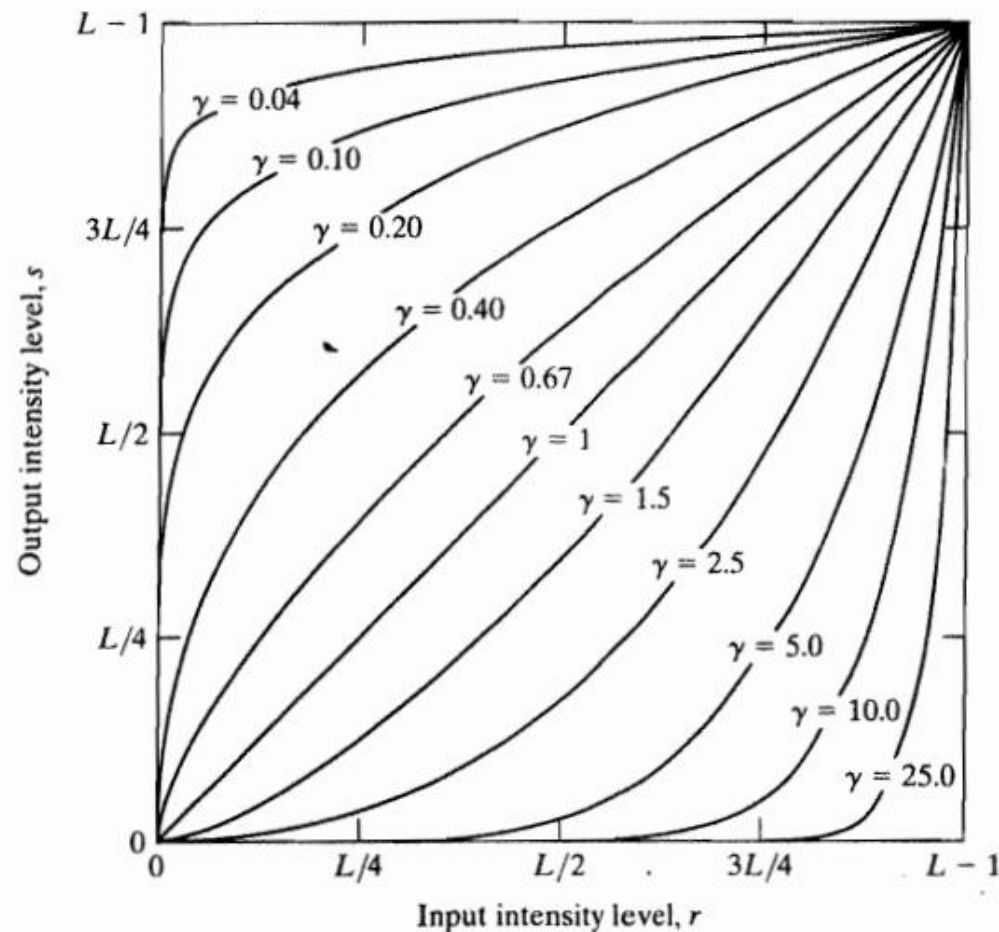
□ Power-Law Transformation:

$$S = c.r^\gamma$$

If $\gamma = 1$ then the operation perform the image identity.

If $\gamma > 1$, then the contrast of high-value portion is increased at the cost of low value region.

If $\gamma < 1$, then the contrast of high value region is reduced at the cost of low-value region.



Intensity Transformation

❑ **Power-Law Transformation: (Gamma less than 1)**

$$D1 = 1.0 * A^{0.5}$$

General
Equation:

$$S = c.r^\gamma$$

$$C = 1$$

$$\gamma = 0.5$$

Original Image A

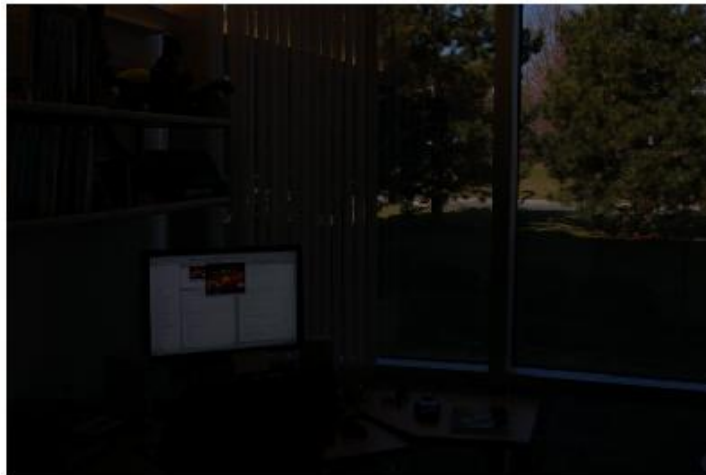
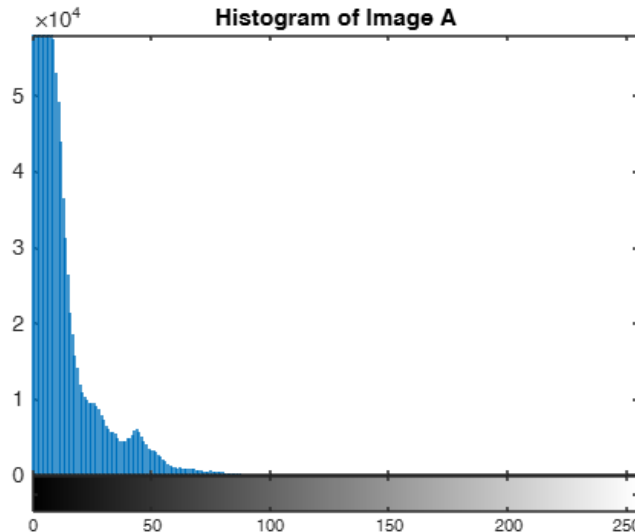


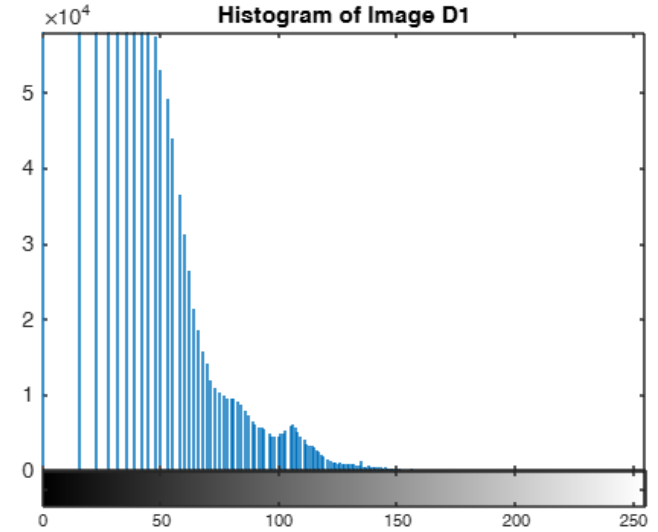
Image D1



Histogram of Image A



Histogram of Image D1



Intensity Transformation

❑ Power-Law Transformation: (Gamma Less than 1)

a b
c d

FIGURE 3.8

(a) Magnetic resonance image (MRI) of a fractured human spine (the region of the fracture is enclosed by the circle).

(b)–(d) Results of applying the transformation in Eq. (3-5) with $c = 1$ and $\gamma = 0.6, 0.4$, and 0.3 , respectively. (Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)



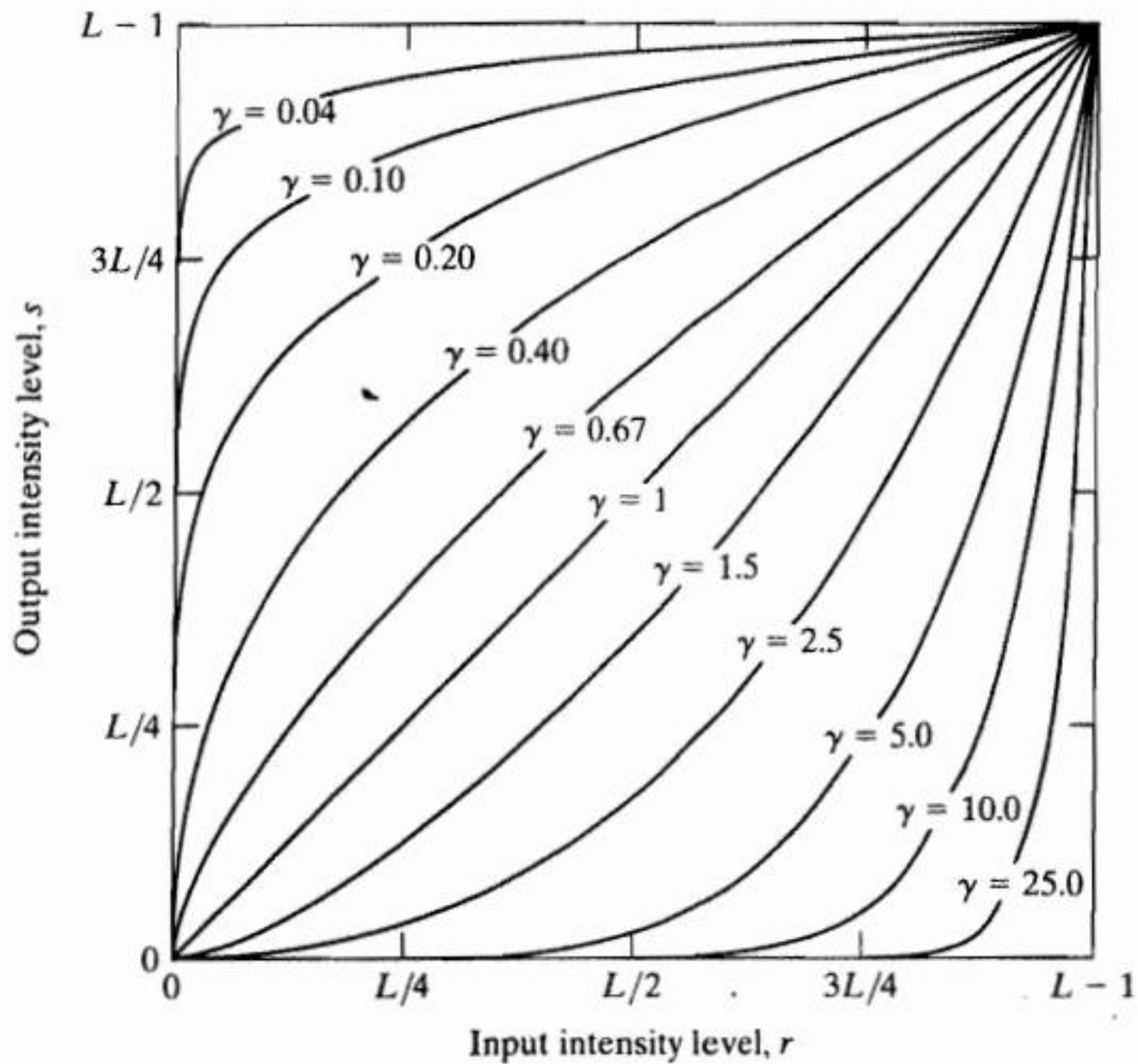
$\gamma=0.6$



$\gamma=0.4$



$\gamma=0.3$



Intensity Transformation

☐ **Power-Law Transformation: (Gamma Greater than 1)**

$$S2 = 1.0 * A^2$$

General Equation:

$$S = c.r^\gamma$$

$$C = 1$$

$$\gamma = 2$$

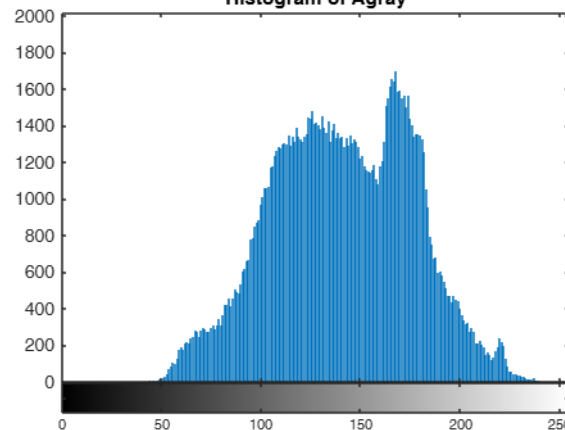
Gray Image Agray



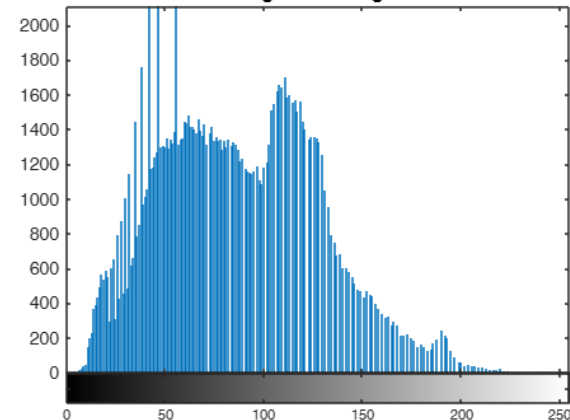
Image S1



Histogram of Agray



Histogram of Image S1



Intensity Transformation

❑ Power-Law Transformation:

a b
c d

FIGURE 3.9

(a) Aerial image.
(b)–(d) Results
of applying the
transformation
in Eq. (3-5) with
 $\gamma = 3.0, 4.0,$ and
 5.0 , respectively.
($c = 1$ in all cases.)
(Original image
courtesy of
NASA.)



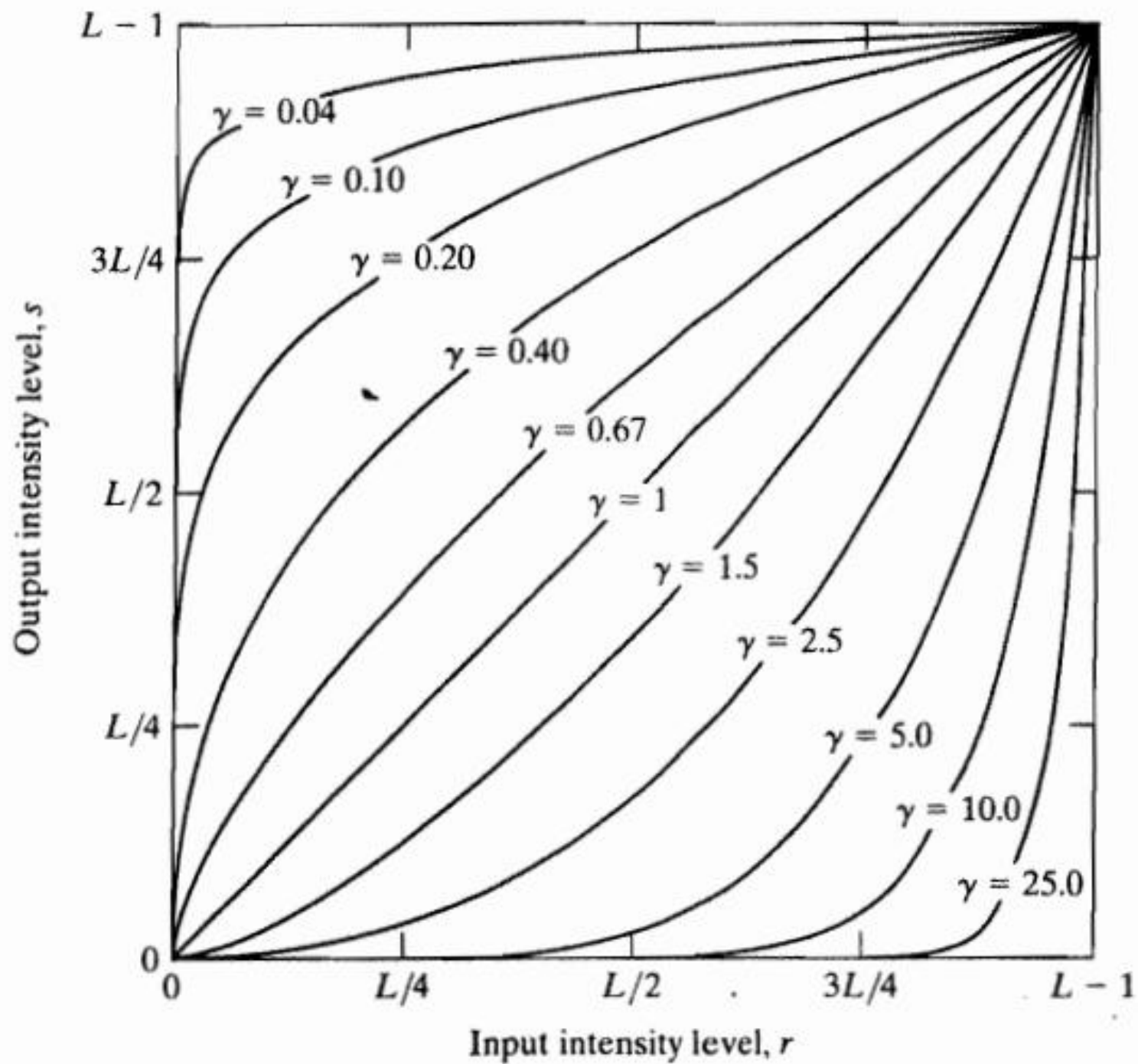
$\gamma = 3.0$



$\gamma = 4.0$



$\gamma = 5.0$



Numerical Examples

Ques 3: Power law transformation $S = c.r^\gamma$ is applied on the image on the left to get the image on the right whose dark regions have become lighter. Is the value of $\gamma > 1$ or < 1 ?

GLA University, Mid term Examination, Odd Semester 2022-23



Numerical Examples

Ques 4: Which of the following is true for Power Law Transformation? Write the complete true statement in your answer sheet.

GLA University, Mid term Examination, Odd Semester 2023-24

- $\gamma > 1$ Compresses dark values, Expands Bright Values
- $\gamma > 1$ Expands dark values, Compresses Bright Values
- $\gamma < 1$ Compresses dark values, Expands Bright Values
- $\gamma = 1$ Expands dark values, Compresses Bright Values

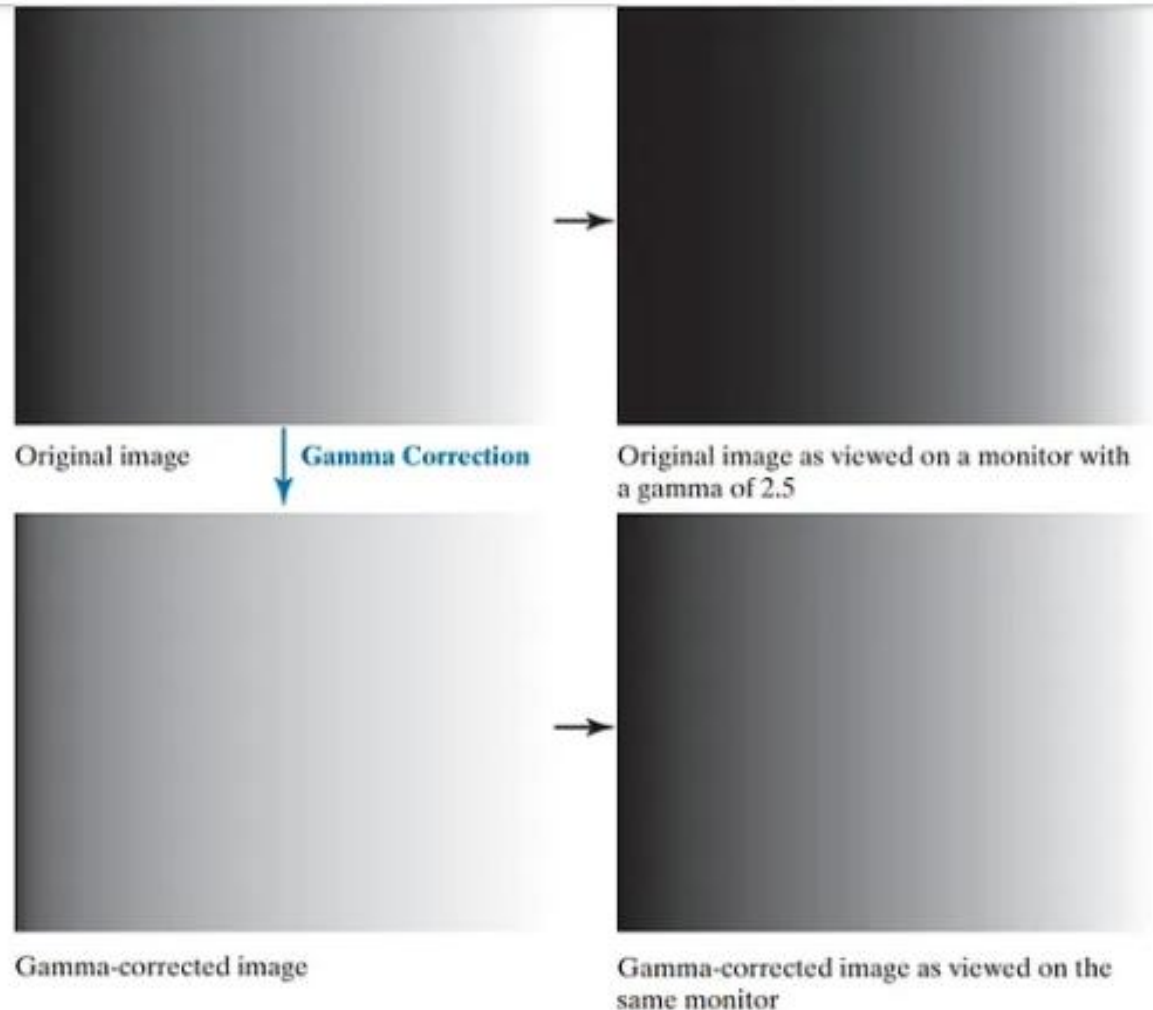
Intensity Transformation

❑ Power-Law Transformation:

a b
c d

FIGURE 3.7

(a) Intensity ramp image.
 (b) Image as viewed on a simulated monitor with a gamma of 2.5.
 (c) Gamma-corrected image.
 (d) Corrected image as viewed on the same monitor. Compare (d) and (a).



Piece-Wise Linear Transformation

Piecewise Linear Transformation

❑ Advantage:

- Quite less complex compared to other image enhancement techniques

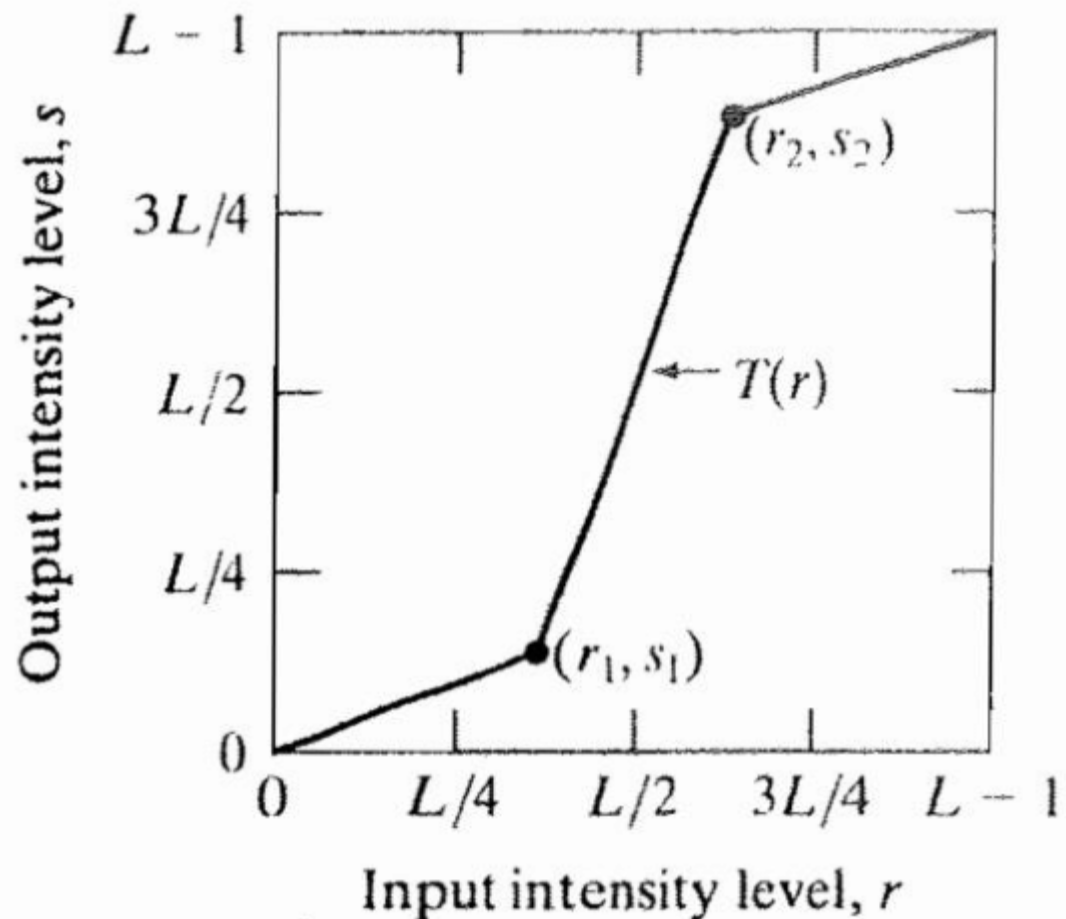
❑ Disadvantage:

- It requires more user inputs.

Piecewise Linear Transformation

□ Classification of Piecewise Linear Transformation Techniques:

- Contrast stretching
- Intensity Level slicing
- Bit plane slicing.



Piecewise Linear Transformation

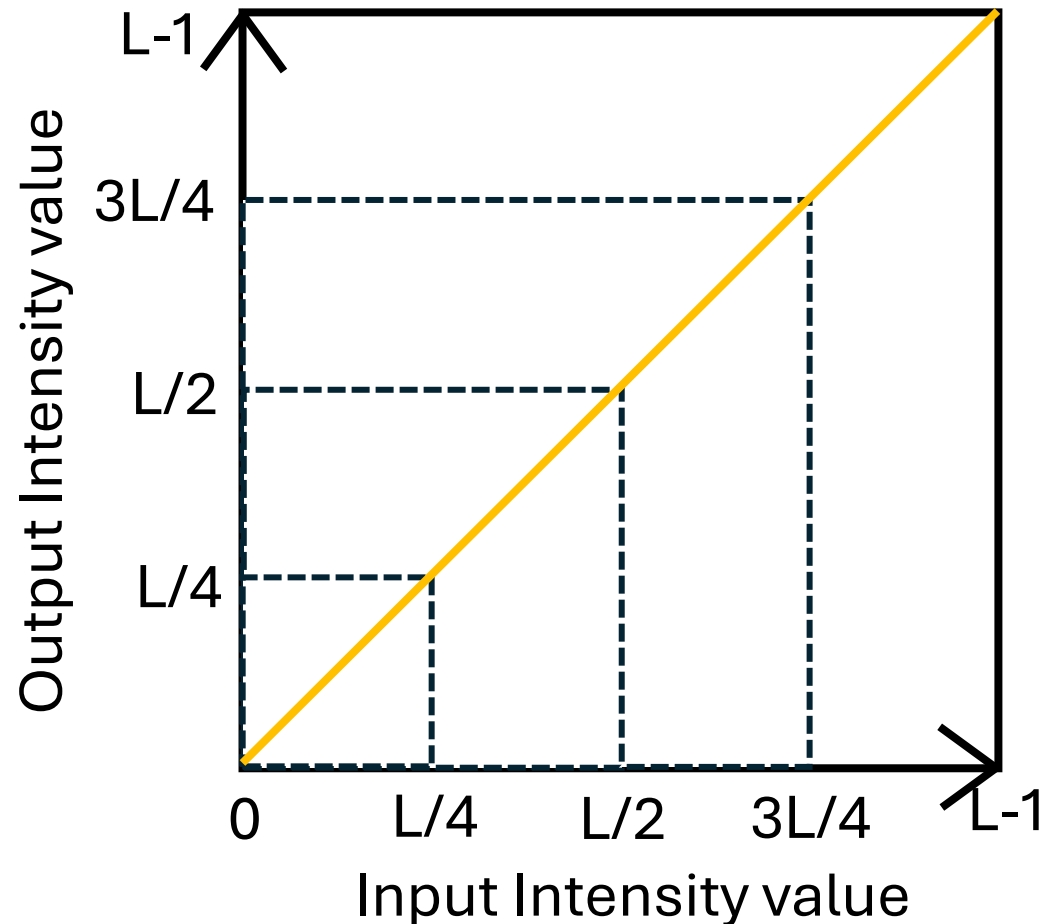
❑ **Case 1: if $r1 = s1$ and $r2 = s2$**

For simplicity:

$$r1 = s1 = L/4$$

$$\text{and } r2 = s2 = 3L/4$$

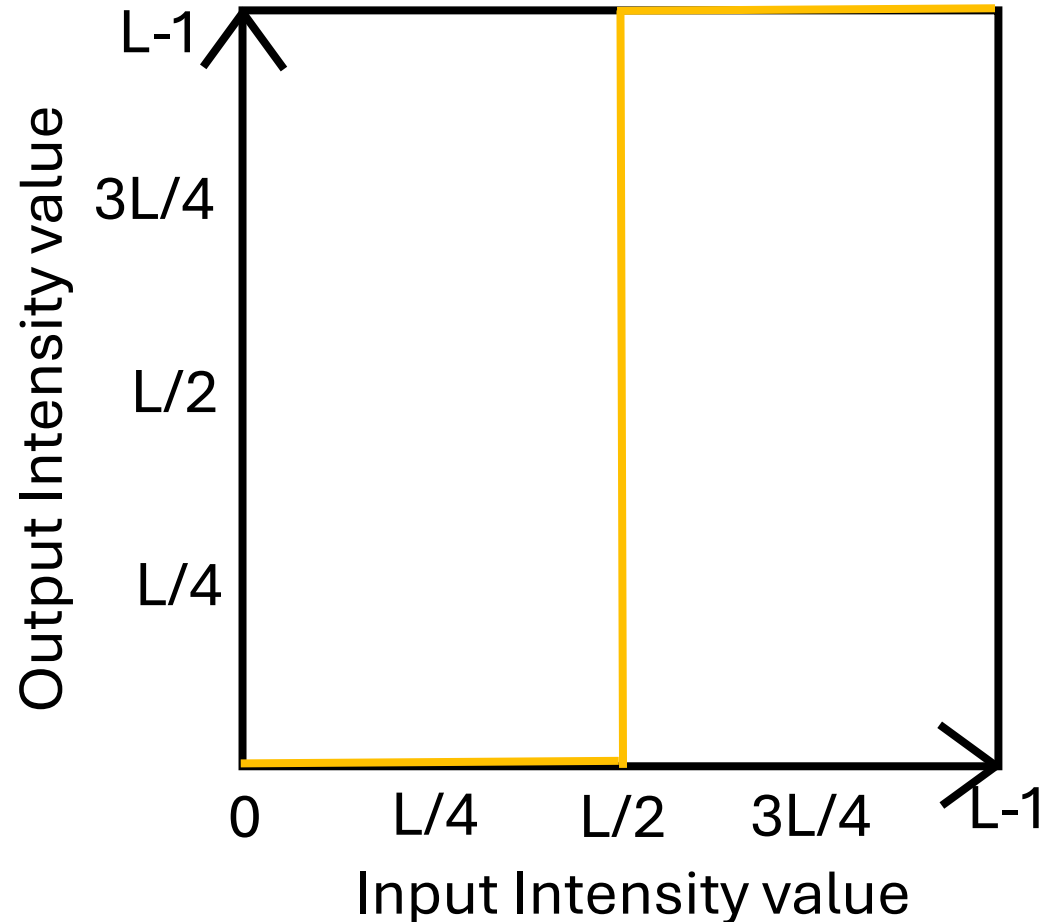
**Image
Identity**



Piecewise Linear Transformation

❑ **Case 2: if $r1 = L/2$, $S1 = 0$ and $r2 = L/2$, $S2 = L-1$**

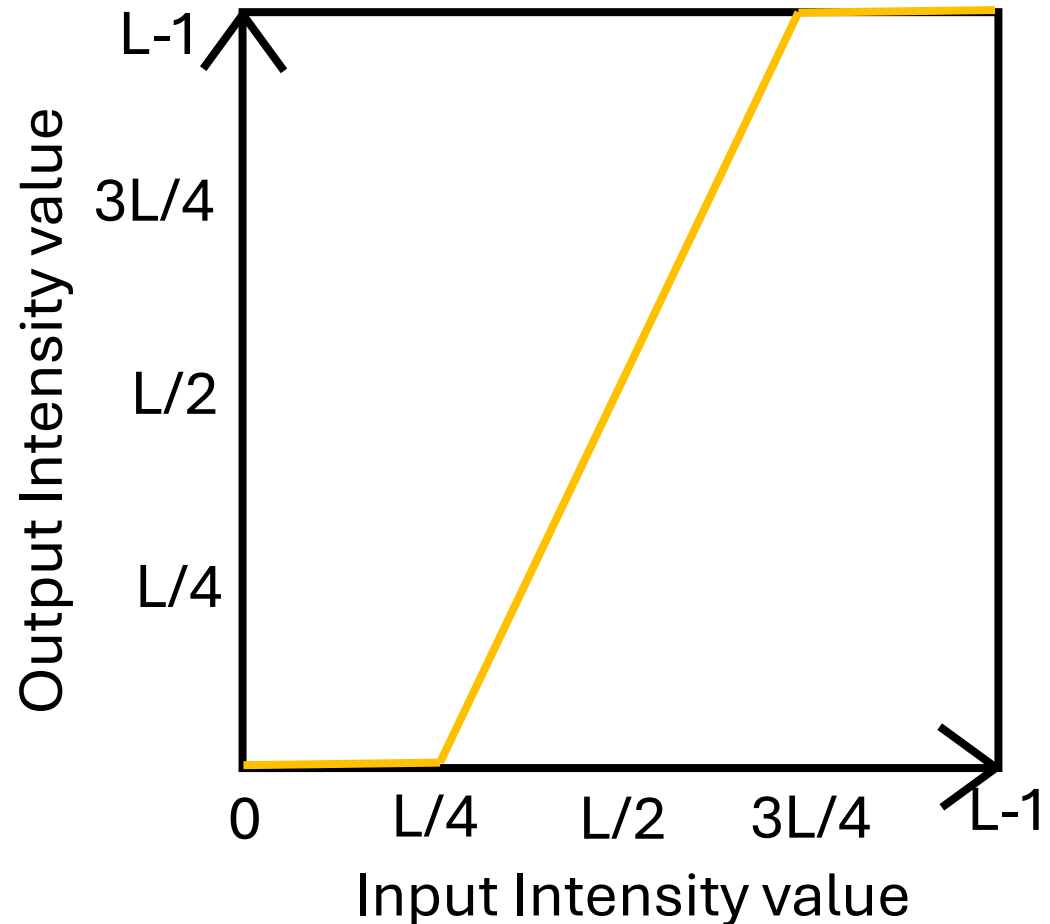
Image
Thresholding



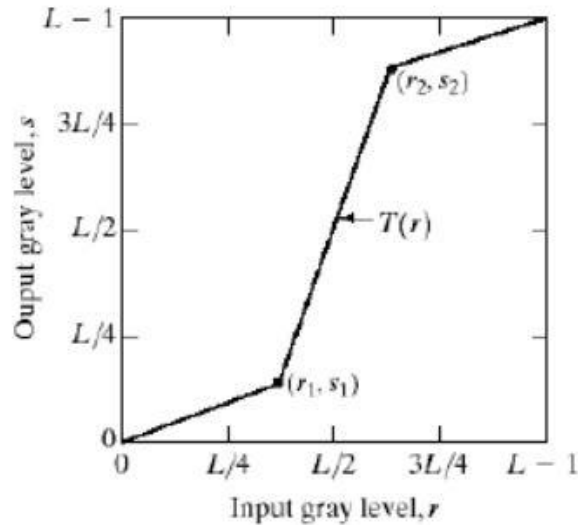
Piecewise Linear Transformation

❑ **Case 3: if $r1 = L/4$, $S1 = 0$ and $r2 = 3L/4$, $S2 = L-1$**

Contrast
Stretching



Piecewise Linear Transformation



a b
 c d

FIGURE 3.10
 Contrast stretching.
 (a) Form of transformation function. (b) A low-contrast image. (c) Result of contrast stretching. (d) Result of thresholding. (Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

Intensity Level Slicing

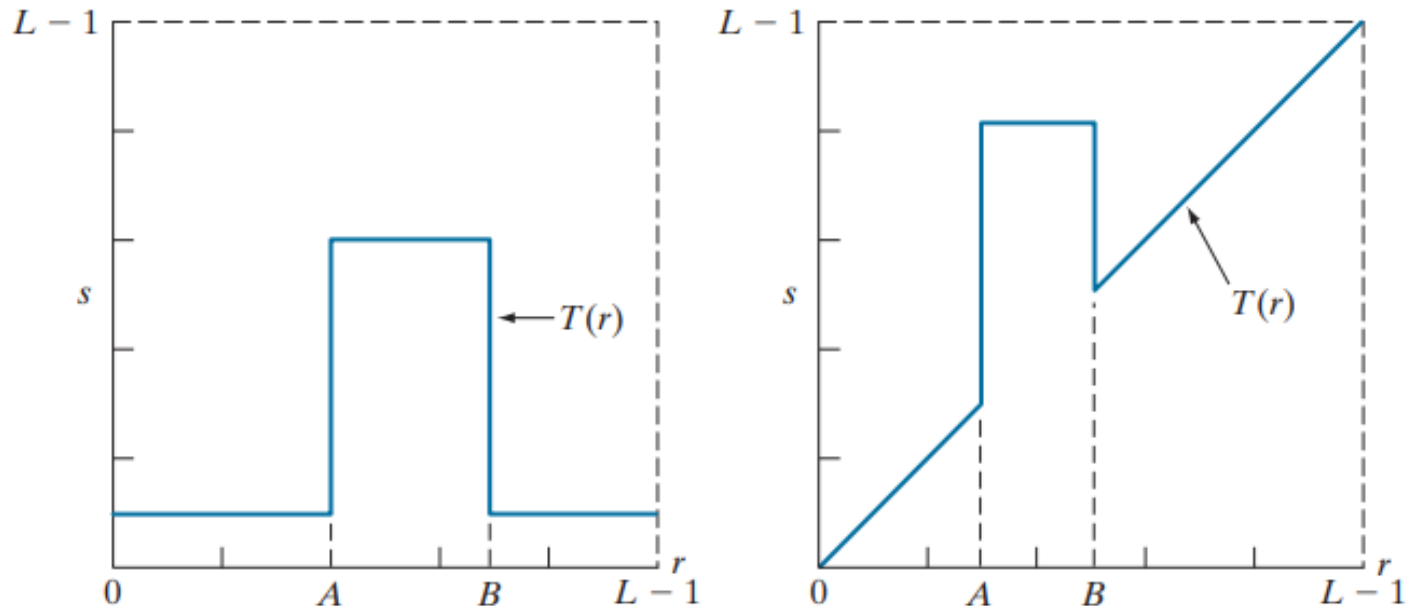
- There are applications in which it is of interest to highlight a specific range of intensities in an image.
- Applications:**
 - Enhancing features in satellite imagery.
 - Enhancing flaws in X-ray images

a b

FIGURE 3.11

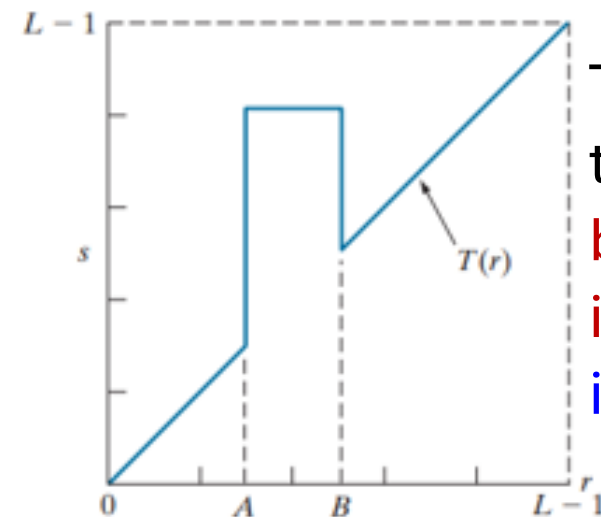
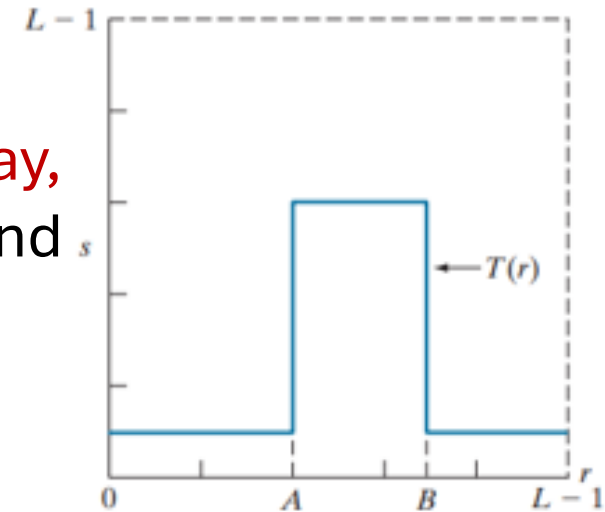
(a) This transformation function highlights range $[A, B]$ and reduces all other intensities to a lower level.

(b) This function highlights range $[A, B]$ and leaves other intensities unchanged.



Intensity Level Slicing

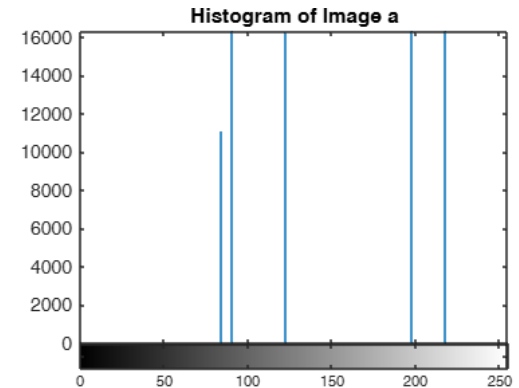
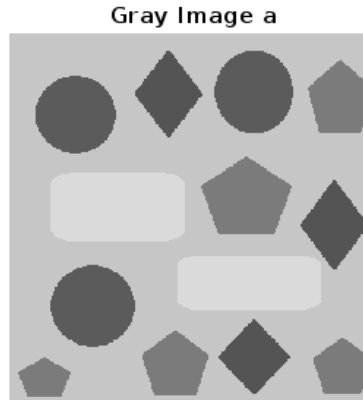
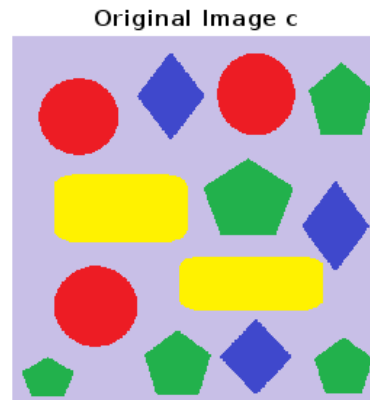
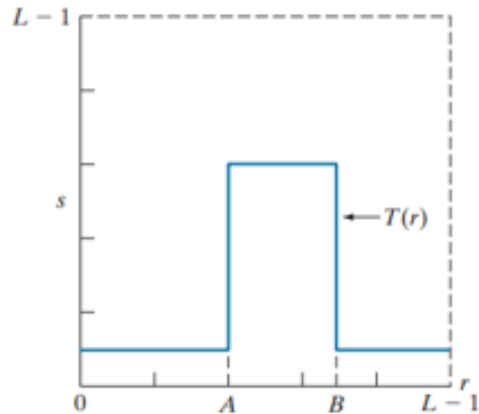
One approach, is to display in **one value (say, white)** all the values in the range of interest and in **another (say, black)** all other intensities.



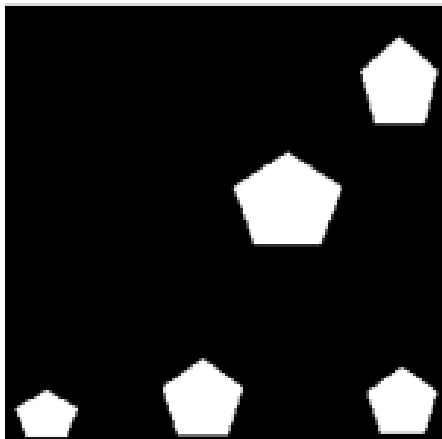
The second approach, based on the transformation curve shown in figure b, **brightness (or darkness) the desired range of intensities**, but leaves **all other intensity levels in the image unchanged**.

Intensity Level Slicing

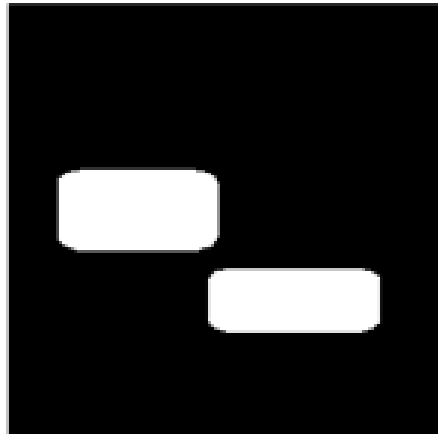
Without Background



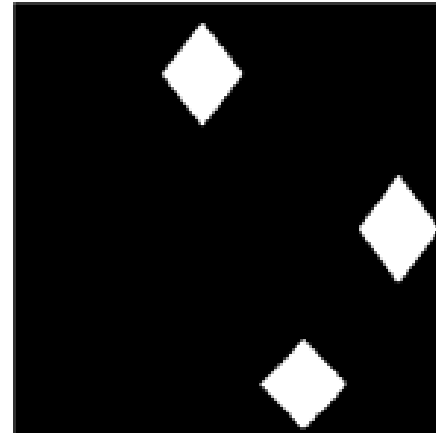
For $r_1 = 100$ and $r_2 = 150$
Output Image b



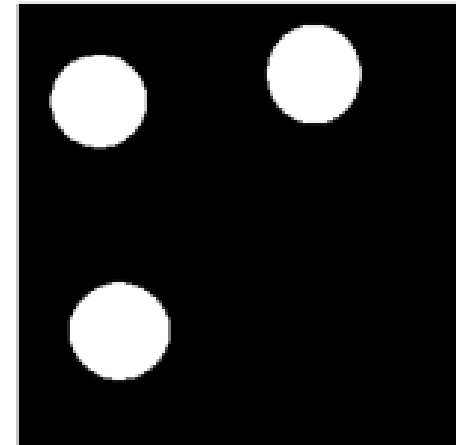
For $r_1 = 200$ and $r_2 = 250$
Output Image b



For $r_1 = 75$ and $r_2 = 85$
Output Image b

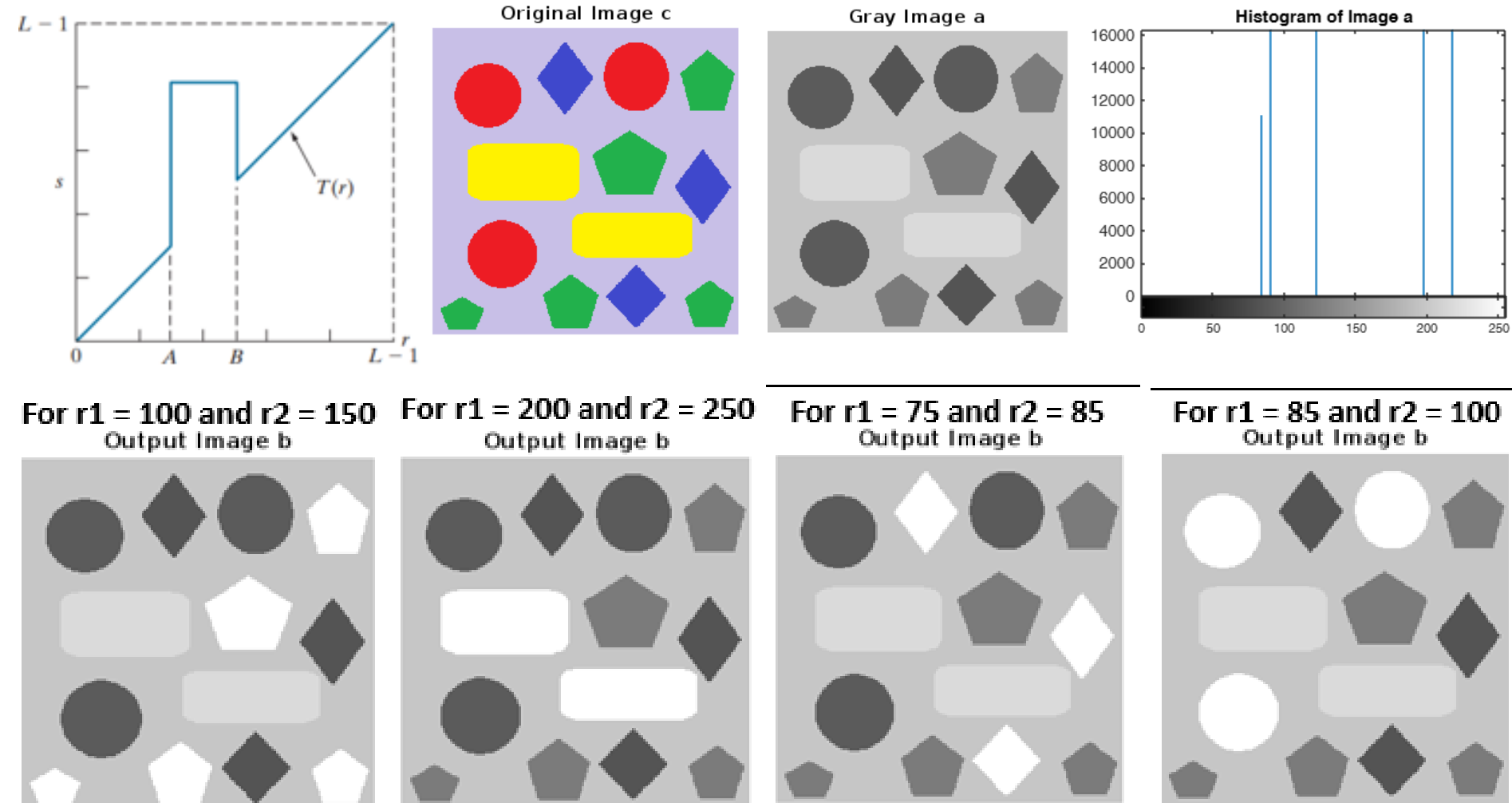


For $r_1 = 85$ and $r_2 = 100$
Output Image b

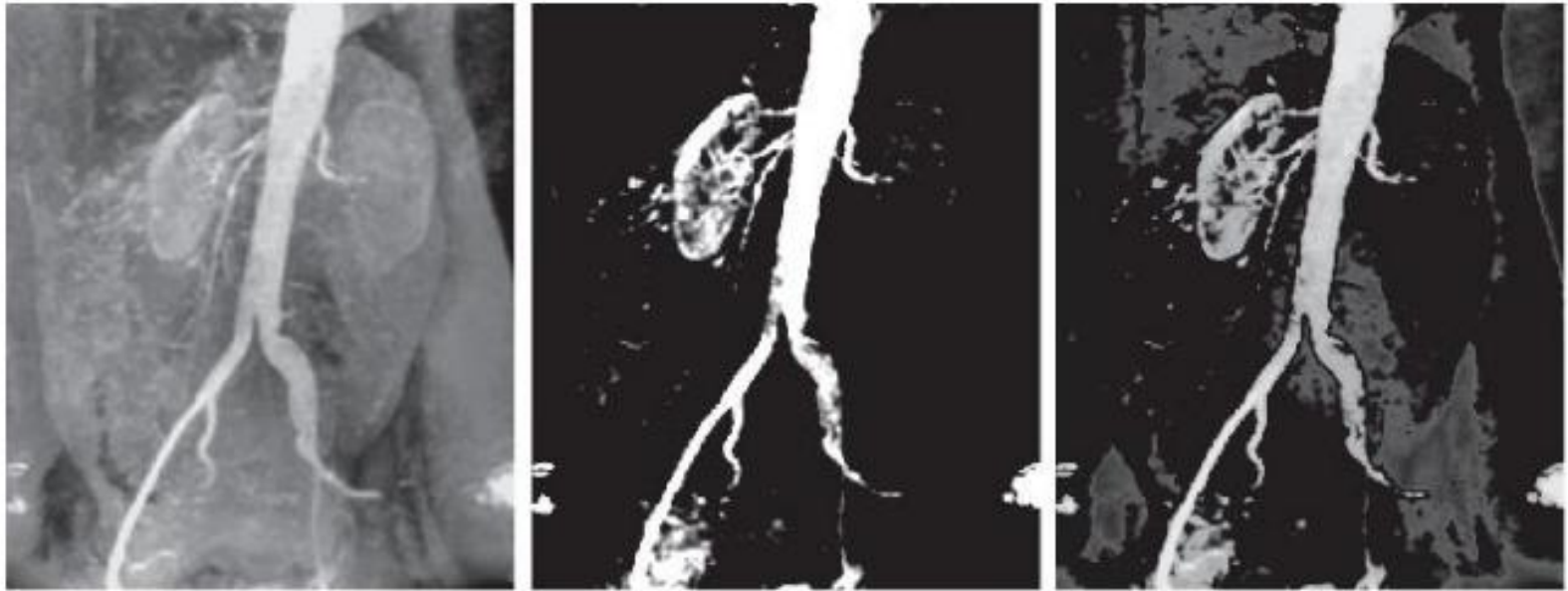


Intensity Level Slicing

With Background



Intensity Level Slicing



a b c

FIGURE 3.12 (a) Aortic angiogram. (b) Result of using a slicing transformation of the type illustrated in Fig. 3.11(a), with the range of intensities of interest selected in the upper end of the gray scale. (c) Result of using the transformation in Fig. 3.11(b), with the selected range set near black, so that the grays in the area of the blood vessels and kidneys were preserved. (Original image courtesy of Dr. Thomas R. Gest, University of Michigan Medical School.)

Numerical Examples

Ques 5: Perform Gray level Slicing, on the following 3-bit image. Let $A=3$, and $B=5$. Give the modified image with and without background transformation in the format given below. Also plot the graphs for both types of transformation.

GLA University, Mid term Examination, Odd Semester 2023-24

2	2	1	1	2
2	3	4	5	2
6	7	6	7	2
2	6	5	6	1
1	0	3	3	1

Bit-Plane Slicing

- Highlighting the contribution made to total image appearance by specific bits
- Useful for analyzing the relative importance played by each bit of the image

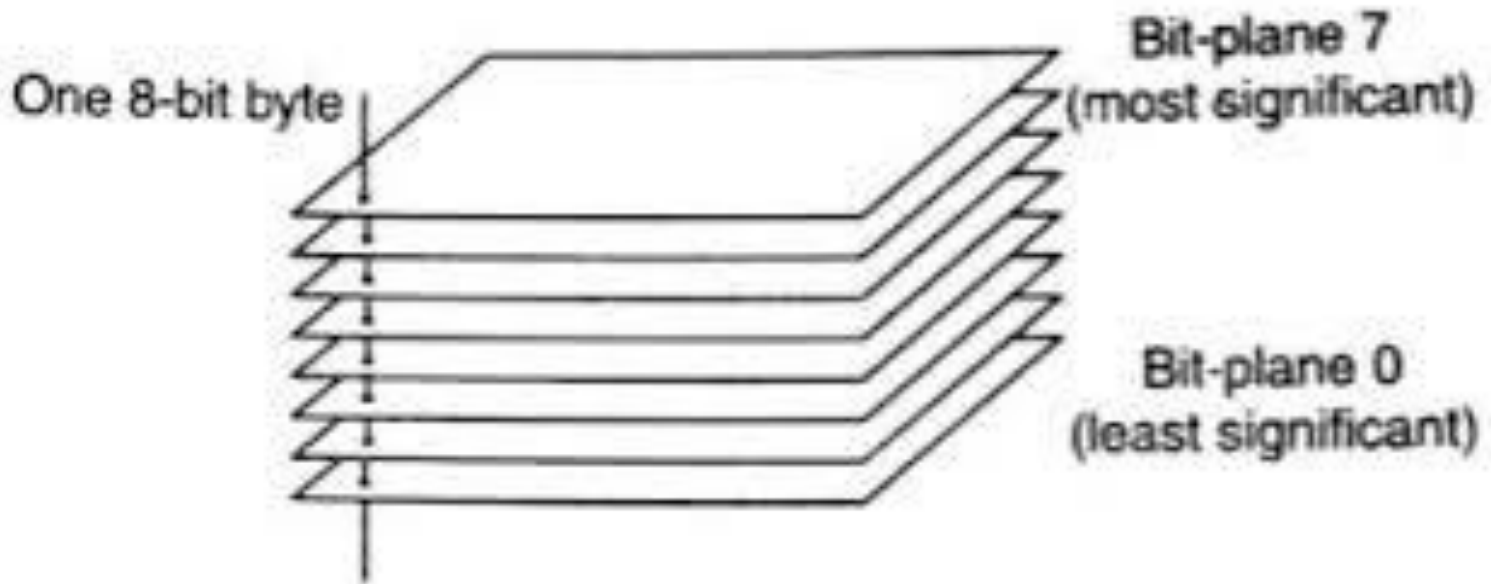


Fig no: 7

Bit-Plane Slicing

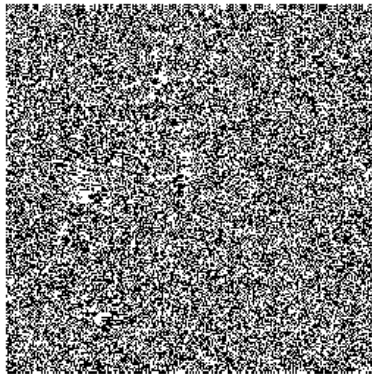
original image



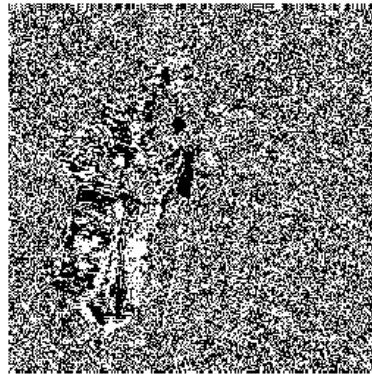
double image



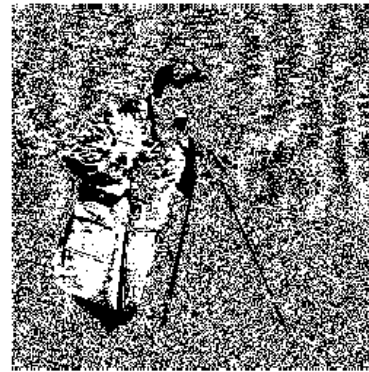
bit plane 0



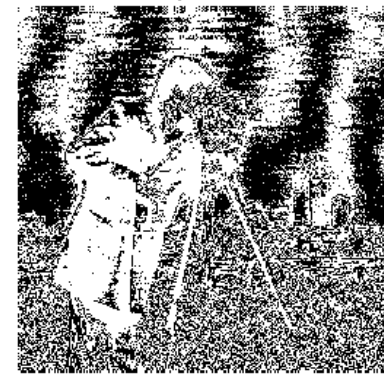
bit plane 1



bit plane 2



bit plane 3



bit plane 4



bit plane 5



bit plane 6



bit plane 7



Bit Plane Slicing



a	b	c
d	e	f
g	h	i

FIGURE 3.14 (a) An 8-bit gray-scale image of size 550×1192 pixels. (b) through (i) Bit planes 8 through 1, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image.

Numerical Examples

Ques 6: consider the following 4-bit, 4×4 image

4	9	1	0
1	2	5	7
5	1	2	15
2	4	6	7

GLA University, Mid term Examination, Even Semester 2021-22

- Extract the 0^{th} bit plane.
- How will the image look if thresholding is set at 3?

Numerical Examples

Ques 7: Consider the following 5-bit, 4 X 4 image

31	30	29	28
16	26	25	21
24	25	26	16
28	29	30	31

**GLA University,
Mid term Examination, Odd
Semester 2024-25**

(i) Extract the 2nd Bit Plane.

(ii) How will image look in binary, if thresholding is set at 27?

Numerical Examples

Ques 8: Consider the following 5 bit, 4×4 image

GLA University, End Sem Exam, 2022-23

A) Extract the first bit plane

	0	1	2	3
0	31	30	29	28
1	16	26	25	24
2	24	25	26	16
3	28	29	30	31

B) How will the image look if thresholding is set at 27? The output image should be still be a 5th bit plane.

C) What will be the new value to the pixel (2,2) if smoothing is done using a 3×3 average filter?

Practice Questions

Numerical Examples

Ques 9: A 7-bit digital image has a histogram where the gray levels are equally distributed in the range from 70 to 110. For each operation of the following transformation functions, describe the gray levels range in which the pixels will lie. Also draw the transformation function for each.

GLA University, Mid term Examination, Even Semester 2022-23

- Image negative
- Addition of 40 to all pixels gray levels
- Application of thresholding functions where the threshold is selected as gray level 95.

Numerical Examples

Ques 10: Obtained the transformed image shown in Fig 1a, by using the transformation function shown in Fig 1 b

GLA University, Carry Exam, 2023-24

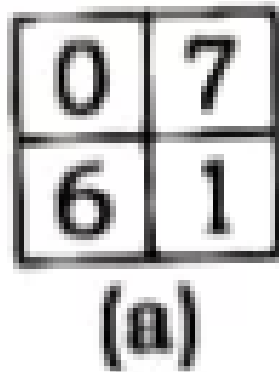


Fig. 1

Numerical Examples

Ques 11: An 8-bit digital image has a histogram where the gray levels are equally distributed in the range of 160 to 220. For each operation of the following transformation, describe the gray level range in which the pixels will lie. Also draw the transformation function for each. In each case, a gray level image will be generated where the gray levels can not be less than 0 or more than 255.

GLA University, Mid term Examination, Even Semester 2021-22

- Image negative
- Addition of 50 to all pixels gray levels.
- Application of thresholding function where the threshold is selected as gray level 128.

Numerical Examples

Ques 12: For the following given image

1	2	3	4
5	5	6	6
6	7	6	6
6	7	2	3

Assuming the gray level is 0 to 7 i.e. 8, determine the following

- Inversion/digital negative operation $g(x, y) = L-1-f(x, y)$
- Square function $g(x, y) = f(x, y)^2$
- Logarithmic operation $g(x, y) = 0.5 * \log[f(x, y)] + 1$
- Power law operation $g(x, y) = 1 * f(x, y)^{1.2}$

Numerical Examples

Ques 13: A student has a grayscale image of a landscape where the darker areas represent shadows, and the lighter areas represent bright regions. The student wants to apply an intensity transformation on the image effectively to reverse the intensity levels so that the shadows become bright, and the bright regions become dark. Explain which transform should he use. Apply the same transformation on the given image.

GLA University, Mid term Examination, Odd Semester 2024-25

122	150	200
225	225	225
250	250	240

Numerical Examples

Ques 14: Obtain the 3rd bit plane for the following image segment as shown below. Further, generate a binary image by apply thresholding at 15 and write the image.

GLA University, Mid Sem Examination, Even Semester 2023-24

20	24	15
30	5	16
2	18	12

Numerical Examples

Ques 15: You have 4×5 image as follows:

2	3	5	5	6
5	3	9	11	12
12	11	1	8	1
7	7	9	0	1

**GLA University,
Mid term Examination,
Odd Semester 2022-23**

Perform Gray level Slicing with respect to given function – clearly mention the new pixel values in 4×5 matrix. Also, Represent the function graphically.

$$S = \begin{cases} 15 & \text{if } 5 < r \leq 10 \\ r & \text{otherwise} \end{cases}$$



Thank You